
EN 605: Modeling and Simulation of Energy Systems

Course Code	EN 605
Course Name	Modeling and Simulation of Energy Systems
L-T-P-C	2-0-2-3
Prerequisites	None
Intended for	UG/PG
Distribution	Compulsory for M. Tech in Energy Engineering and elective for others
Approval	9th Senate (May 2016)

Course Contents

- **Physical Modelling**
Dimensional analysis and dimensionless groups, Similarity criteria, development of physical models based on similarity and dimensionless group. Application of physical modelling to energy systems. [6 Lectures]
- **Mathematical modelling**
Continuous time dynamic modelling through differential equation, lumped parameter and distributed parameter models, Input-output models, transfer function, state space models and their transfer functions, models for control, transient and steady state response, stability; Simulation of models by MATLAB, applications of dynamic modelling and simulation to energy management in ships and PV systems. [8 Lectures]
- **Fuzzy logic Modelling**
Fuzzy sets, membership function, operations with fuzzy sets, Fuzzy knowledge, Fuzzy inference systems, fuzzy approach to identify power transformer criticality, data clustering to evolve fuzzy models, design of fuzzy logic controller for optimal control of boiler-turbine system. [5 Lectures]
- **Artificial neural network (ANN) modelling**
Neuron, Neural network and its architecture, training, validation and test of the back propagation network (BPN), modelling of solar distillation system, ANFIS models. [6 Lectures]
- **Non-traditional techniques of optimization**
simulated annealing and genetic algorithm. [3 Lectures]

Text Books

1. Josef Kunes, Similarity and Modelilng in Science and Engineering, Cambridge International Science Publishing, 2012.
2. Katsuhiko Ogata, Modern Control Engineering, Prentice Hall India,
3. Timothy J. Ross, Fuzzy logic with eEngineering Applications, John Wiley & Sons, 2010.
4. Simon Haykin, Neural Networks, Pearson Education Inc.,1999.
5. K. Deb, Optimization for Engineering Design, Prentice Hall India Pvt. Ltd.,1991.